A
Requirement Analysis Report

For

Handwritten Digit Recognition

is submitted in partial fulfillment of the requirement for the award of degree of

Bachelor of Technology

(VI Semester B. Tech. for the course **Basics of Machine & Deep Learning** (**PCCAD602T**)) in

Artificial Intelligence & Data Science

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Problem Statement: Handwritten Digit Recognition

Objectives:

- Develop a reliable system to recognize and classify handwritten digits (0–9).
- Implement and compare machine learning and deep learning models for accuracy and efficiency.
- Train models to generalize well across diverse handwriting styles.
- Achieve high recognition accuracy using techniques like Convolutional Neural Networks (CNNs).
- Provide a user-friendly interface for real-time handwritten digit recognition.

Introduction:

Artificial intelligence (AI) in simple words is basically making a computer do the work that traditionally requires the human brain. AI has the ability to take in large amounts of data unlike the human and uses that data to recognize patterns, make decisions, and give judgment. In this AI we have a subset which is called Machine learning. ML is used to make computers to learn to behave as humans. This is done by two ways, supervised learning in which the computer is given a set of input data and the required output for it. Now it uses ML to learn the algorithm to understand how that particular input gives this particular output. Now the unsupervised learning is when input data is provided but with no output, so the ML has to learn to analyze and clutter the datasets into categories.

It's a network architecture for deep learning. It learns from the data which is through the input images. It finds patterns in images and recognizes objects and categories, this is feature extraction. Now max pooling and convolution takes place. Handwritten digit recognition is a classic problem in computer vision and machine learning where the goal is to accurately classify images of handwritten digits (0–9). This problem serves as a benchmark for evaluating the performance of machine learning and deep learning algorithms.

Detection of handwritten numbers, including accuracy in these areas, has reached human perfection using deep convolutional neural networks (CNNs). Recently CNN has become one of the most attractive approaches and has been the ultimate factor in recent success and in several challenging machine learning applications. Considering all the factors stated above have chosen CNN for our challenging tasks of image classification. Used it to identify handwritten numbers, which is one of the higher education and business transactions.

To tackle these challenges, this project focuses on developing a **robust handwritten digit recognition model** using **Convolutional Neural Networks (CNNs) and Residual Networks (ResNet)**.

Importance of Handwritten Digit Recognition

- Widely used in **postal services, banking (cheque processing), and document digitization**.
- Plays a crucial role in **automated form processing and AI-driven applications**.
- Serves as a benchmark for evaluating machine learning and deep learning models.

Challenges in Handwritten Digit Recognition

- **Diverse handwriting styles** → Different individuals write in unique ways.
- **Size and orientation variations** → Digits can be skewed, rotated, or stretched.
- **Noisy and low-quality images** → Blurred or distorted inputs affect accuracy.
- **Data complexity** → Digits like '3' and '8' or '1' and '7' can be difficult to distinguish.

Deep Learning Approaches: CNNs and ResNet

a) Convolutional Neural Networks (CNNs)

- CNNs are highly efficient for image recognition and feature extraction.
- **Key CNN components**:
  - **Convolution layers** → Detect patterns like edges, curves, and textures.
  - **Pooling layers** → Reduce dimensionality while preserving important features.
  - **Fully connected layers** → Classify the extracted features into digit categories.

b) Residual Networks (ResNet)

- ResNet introduces **skip connections** to solve the **vanishing gradient problem** in deep networks.
- **Key benefits of ResNet**:
  - Allows training of **ultra-deep networks** without performance degradation.
  - Learns intricate patterns, improving accuracy on **noisy and distorted data**.
  - Outperforms traditional CNNs in recognizing complex handwriting variations.

Dataset and Implementation

- **Dataset Used → MNIST (Modified National Institute of Standards and Technology)** dataset, containing **60,000 training** and **10,000 testing** images of handwritten digits.
- **Implementation Goals**:
  - Develop a **CNN-based model** for digit classification.
  - Implement **ResNet** to enhance feature extraction and accuracy.
  - Optimize the model for **high computational efficiency**.

Dataset:

https://www.kaggle.com/datasets/hojjatk/mnist-dataset

curl -O http://yann.lecun.com/exdb/mnist/train-images-idx3-ubyte.gz

curl -O http://yann.lecun.com/exdb/mnist/train-labels-idx1-ubyte.gz

curl -O http://yann.lecun.com/exdb/mnist/t10k-images-idx3-ubyte.gz

curl -O http://yann.lecun.com/exdb/mnist/t10k-labels-idx1-ubyte.gz

The provided links are the official source for downloading the **MNIST (Modified National Institute of Standards and Technology)** dataset, which is widely used for handwritten digit recognition tasks.

Description of the Links:

1. **train-images-idx3-ubyte.gz**
   - This file contains the training images in compressed format.
   - It includes 60,000 grayscale images of handwritten digits, each with a resolution of 28x28 pixels.
2. **train-labels-idx1-ubyte.gz**
   - This file contains the corresponding labels for the training images.
   - Each label represents the digit (0–9) depicted in the corresponding training image.
3. **t10k-images-idx3-ubyte.gz**
   - This file contains the test images in compressed format.
   - It includes 10,000 grayscale images of handwritten digits, also with a resolution of 28x28 pixels.
4. **t10k-labels-idx1-ubyte.gz**
   - This file contains the corresponding labels for the test images.

- Each label represents the digit (0–9) depicted in the corresponding test image.

Literature Survey:

https://sist.sathyabama.ac.in/sist_naac/documents/1.3.4/1822-b.e-ece-batchno-14.pdf

https://ijsrcseit.com/paper/CSEIT217536.pdf

https://www.researchgate.net/publication/347943261_Handwritten_Digit_Recognition_Using_Machine_Learning

https://osf.io/5sc7d/download

https://www.researchsquare.com/article/rs-2038607/v1.pdf

https://www.rgmcet.edu.in/NAAC/2023/1.3.3/Project%20Reports/2021-22-Projects/2021-22-CSE-Handwritten%20Digit%20Recognition-4.pdf

https://www.scitechnol.com/peer-review/handwritten-digit-recognition-qoiv.php?article_id=21218

https://rljit.in/naacssr/criterion-1/1.3.2/Project_Intern_Certificates/130.pdf

https://ijcsmc.com/docs/papers/January2022/V11I1202214.pdf

https://www.irjmets.com/uploadedfiles/paper//issue_3_march_2022/19432/final/fin_irjmets1646225071.pdf

Conclusion:

Handwritten digit recognition using machine learning and deep learning demonstrates the capability of algorithms to interpret and classify complex visual patterns. By leveraging the MNIST dataset, which serves as a benchmark in the field of computer vision, we can develop robust models that achieve high accuracy across various handwriting styles. This project not only provides insights into model performance but also showcases the practical applications of machine learning and deep learning in tasks like automated form processing, postal code recognition, and document digitization. The study contributes to advancing technologies that enhance human-machine interaction and automate routine tasks effectively.